## Approximating the exchange-correlation energy

Since the exact expression for the exact exchange-correlation functional remains unknown, we must resort to approximations. The simplest one is the local density approximation (LDA). This approximation is based on the observation that it is possible to solve the many-electron SE in certain situations. In particular, one can use a technique called Quantum Monte Carlo to obtain accurate solutions for a many-electron system in free space, i.e. without atomic nuclei. In this case, we have $V(\mathbf{r}) = 0$ and as a consequence the ground state density must be constant everywhere, i.e. $n(\mathbf{r}) = n$. By solving the many-electron SE, we can calculate the ground state energy $E_0$ per unit volume as function of $n$. From this we can subtract the kinetic energy density and the Hartree energy per unit volume to determine the exchange-correlation energy per unit volume for electrons in free space. We will denote this quantity by $\epsilon_{xc}^{FS}(n)$.
Of course, in a real material there are atomic nuclei and therefore the ground state density of electrons $n(\mathbf{r})$ will not be a constant, but inhomogeneous in space. The idea of the local density approximation is that we can approximate the total exchange-correlation energy as a sum of contributions from each point in space and that the contribution from position $\mathbf{r}$ is given by $dV\epsilon_{xc}^{FS}(n(\mathbf{r}))$. In other words, we assume that the electrons at position $\mathbf{r}$ are behave exactly as if they were in free space and embedded in an environment where the density is the same as the one at $\mathbf{r}$. The total exchange-correlation energy can then be expressed as

$$E_{xc}^{LDA}[n] = \int dV \epsilon_{xc}^{FS}(n(\mathbf{r})). \tag{1}$$

Besides the LDA, there exists a zoo of other approximations to $E_{xc}$. For example, the class of generalized gradient approximations (which include the widely used PBE functional named after its creators Perdew, Becke and Ernzerhof) also depend on the gradient of the density, i.e. $E_{xc}[n, \nabla n]$. This makes it possible to take into account the dependence of the exchange-correlation energy on the local environment. In recent years, hybrid exchange-correlation functionals have become very popular. These include the HSE functional (named after Heyd, Scuseria and Ernzerhof) or the PBE0 functional (an extension of the PBE functional). These functionals incorporate the environment dependence of the exchange-correlation energy more explicitly, but at the cost of being computationally more demanding. Finally, there has been much work on including the effect of van der Waals interactions in the exchange-correlation energy. Such van der Waals functionals must be used to describe layered materials, organic crystals or biomaterials.
Choosing the appropriate exchange-correlation functional for a specific application requires some experience. The LDA and the generalized gradient approximations (GGAs) are often very good for calculating structural, elastic and vibrational properties of materials. However, they are often inaccurate for atomization energies, ionization potentials, band gaps of solids and activation energies of chemical reactions. For these properties, it is often necessary to use hybrid functionals.

## Solving the Kohn-Sam equation on a computer

Assume that we want to solve the Kohn-Sham equations for a molecule, say $N_2$. As discussed in the second lecture, we have to carry out a self-consistent procedure where we start from an initial density and then proceed to construct a corresponding KS potential which we then use to solve the KS equations and determine new KS wavefunctions from which an improved approximation to the ground state density can be obtained. In this section, we will discuss how - for a given KS potential - we can obtain the corresponding solutions of the KS equations.
As a first step we will invoke periodic boundary conditions (note that this is not a necessity, but this is what Quantum Espresso does). This means that instead of considering a single molecule, we consider an infinite array of periodically repeated molecules. We will assume that the unit cell of this system (which contains a single molecule) is a cube with side length $a$. The KS wavefunctions in every unit cell must be the same and therefore they must be periodic functions which can be expressed as a sum of sines and

cosines - or alternatively, as a sum of plane waves:

$$\psi(\mathbf{r}) = \sum_{\mathbf{G}} c_{\mathbf{G}} e^{i\mathbf{G}\cdot\mathbf{r}}, \tag{2}$$

where **G** denotes a set of wavevectors which are determined by the periodic boundary conditions. For example, we must have

$$\psi(\mathbf{r} + a\hat{x}) = \psi(\mathbf{r}), \tag{3}$$

which results in $G_x = 2\pi/a \times$integer (and similar equations for the y- and z-components). Moreover, the coefficients $c_{\mathbf{G}}$ are given by

$$c_{\mathbf{G}} = \frac{1}{a^3} \int_{cube} dV \psi(\mathbf{r}) e^{-i\mathbf{G}\cdot\mathbf{r}}. \tag{4}$$

Of course, the KS potential is also the same in every unit cell and therefore can be written as a Fourier sum

$$V_{KS}(\mathbf{r}) = \sum_{\mathbf{G}} V_{KS,\mathbf{G}} e^{i\mathbf{G}\cdot\mathbf{r}}. \tag{5}$$

Inserting the expressions for the KS wavefunction and the KS potential into the KS equations yields

$$\sum_{\mathbf{G}} \frac{\hbar^2\mathbf{G}^2}{2m} c_{\mathbf{G}} e^{i\mathbf{G}\cdot\mathbf{r}} + \sum_{\mathbf{G}_1,\mathbf{G}'} V_{KS,\mathbf{G}_1} c_{\mathbf{G}'} e^{i(\mathbf{G}_1+\mathbf{G}')\cdot\mathbf{r}} = \epsilon \sum_{\mathbf{G}} c_{\mathbf{G}} e^{i\mathbf{G}\cdot\mathbf{r}}. \tag{6}$$

In the second term, we change variables and introduce $\mathbf{G} = \mathbf{G}_1 + \mathbf{G}'$ to get rid of the variable $\mathbf{G}_1$. Rearranging the equation and collecting terms yields

$$\sum_{\mathbf{G}} e^{i\mathbf{G}\cdot\mathbf{r}} \left[ \left\{ \frac{\hbar^2\mathbf{G}^2}{2m} - \epsilon \right\} c_{\mathbf{G}} + \sum_{\mathbf{G}'} V_{KS,\mathbf{G}-\mathbf{G}'} c_{\mathbf{G}'} \right] = 0. \tag{7}$$

As the plane waves are a orthogonal set of functions, the left hand side can only be equal to zero if each coefficient that multiplies the complex exponential vanishes separately. This means that for each wavevector **G**, we have

$$\left\{ \frac{\hbar^2\mathbf{G}^2}{2m} - \epsilon \right\} c_{\mathbf{G}} + \sum_{\mathbf{G}'} V_{KS,\mathbf{G}-\mathbf{G}'} c_{\mathbf{G}'} = 0. \tag{8}$$

This infinite set of equations is equivalent to the KS equations. Note that these equation can be written in matrix form according to $H_{KS}\vec{c} = \epsilon\vec{c}$ where $H_{KS}$ is an infinite dimensional matrix. To make this equation solvable by a computer, we have to truncate the size of the matrix. This can be achieved by only allowing **G** vectors whose length is shorter than a specified cutoff. This condition is usually expressed as

$$\frac{\hbar^2\mathbf{G}^2}{2m} < E_{cut}, \tag{9}$$

where $E_{cut}$ is the so-called plane-wave energy cutoff. The cutoff energy is a convergence parameter. This means you have to choose it sufficiently large to ensure an accurate description of the system under consideration accurately. In practical DFT calculations it is very important to check for convergence, e.g. by verifying that the quantity you are calculating does not change its value as you increase $E_{cut}$.

## Pseudopotentials

The electrons of an atom can often be classified as valence electrons which participate in chemical bonding and core electron which do not. It is therefore often acceptable to only consider the valence electrons and combine the core electrons and the atomic nucleus into an effective nucleus that interacts with the valence electrons through a so-called pseudopotential. The advantage of using pseudopotentials is that one has to deal with fewer electrons and also it is possible to use a lower energy cutoff. This is because representing the strongly localized KS wavefunctions of core electrons requires lots of plane waves and therefore a high cutoff energy. However, pseudopotentials must be used with care: the construction of

accurate pseudopotentials is quite difficult and using a bad pseudopotentials leads to bad results. It is therefore always advisable to first establish the accuracy of a pseudopotential by comparing to all-electron results, i.e. results of DFT calculations that take the core electrons into account explicitly (note that these codes typically do not use plane waves to represent the KS wavefunctions).
Pseudopotentials are constructed using the following procedure. One first solves the KS equation for a single atom including both core and valence electrons. This yields all-electron (AE) wavefunctions $\psi_{AE}(\mathbf{r})$ and energies $\epsilon_{AE}$ for the valence states. Next, we can solve the KS equations for the atom, but using a pseudopotential to describe the core electrons and the atomic nucleus. We impose that the energies $\epsilon_{PP}$ of the valence states that result from this pseudopotential calculation must be equal to the all-electron energies, i.e. $\epsilon_{PP} = \epsilon_{AE}$. Moreover, the so-called pseudowavefunctions $\psi_{PP}(\mathbf{r})$ must be equal to $\psi_{AE}(\mathbf{r})$ in the region where the chemical bonding happens. However, the pseudowavefunction can be different from the all-electron wavefunction near the position of the nucleus. These conditions (together with a recipe for constructing the pseudowavefunction near the atomic nucleus) can be used to construct a pseudopotential for the atom. This pseudopotential can then be used in calculations for molecules and crystals.